

ADITI CHANDRA

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PROFESSIONAL SUMMARY

Robotics & Automation undergraduate with strong hands-on experience in **ROS2, robotics software engineering, Python, embedded systems, autonomous robotics, and computer vision**. Skilled in building intelligent robotic systems using **ROS2, Python, C, OpenCV, Raspberry Pi, ESP32, Linux, Gazebo, and RViz**. Experienced in sensor integration, robot control systems, AI-based automation, and real-time robotic applications. Passionate about developing scalable robotics software solutions for autonomous and remote robotic platforms.

TECHNICAL SKILLS

Programming	Python, C, SQL, Scilab
Robotics	ROS2, Gazebo, RViz, Autonomous Robotics, Robot Simulation, Sensor Integration
Embedded Systems	Raspberry Pi, ESP32, Arduino, Motor Drivers, IoT Systems, Serial Communication
AI & Vision	OpenCV, TensorFlow, CNN, Image Processing, Machine Learning
Tools	Linux, Git, GitHub, Fusion 360

EDUCATION

The Neotia University | B.Tech – Robotics & Automation | 2023 – 2027

CGPA: 9.04

INTERNSHIP EXPERIENCE

Embedded Systems & IoT Intern — Jadavpur University | Jun 2025 – Jul 2025
Developed embedded and IoT-based automation systems using sensors and microcontrollers. Worked on real-time hardware interfacing, communication protocols, and robotics workflows. Integrated sensors and automation modules in Linux-based embedded environments.

PROJECTS

SARTHI — Smart Agriculture Rover

- Developed an autonomous agriculture rover using Raspberry Pi and AI Camera for crop monitoring.
- Implemented plant detection, leaf color analysis, and field analysis using OpenCV and computer vision.
- Integrated robotics control logic, sensors, and embedded hardware modules.

Tech Stack: ROS2, Python, Raspberry Pi, OpenCV, Embedded Systems

Rakshak — Multi-Terrain Amphibious Surveillance Rover

- Built an ESP32 and ESP32-CAM based surveillance rover for disaster monitoring applications.
- Developed a Python-based monitoring dashboard integrated with SQL database systems.
- Implemented human detection and remote monitoring workflows using OpenCV.

Tech Stack: Python, ESP32, SQL, OpenCV, Embedded Robotics

Crop Disease Detection Using CNN

- Developed a CNN-based crop disease detection system using TensorFlow and plant datasets.
- Built image preprocessing and classification pipelines using OpenCV and deep learning models.

Tech Stack: Python, TensorFlow, CNN, OpenCV, Machine Learning

ACHIEVEMENTS

- 1st Position — IEEE R10 Robotics Competition 2025
- 1st Position — Roborace, Srijaan 2026, Jadavpur University
- 1st Position & Overall Winner — AGRINOVA National Hackathon
- 2nd Position — IEM Tech Fest Roborace

WORKSHOPS & TRAINING

- Smart Automation through IoT — AILABS
- Autodesk Inventor Workshop — The Neotia University
- Drone Design & Flight Operations — Inspecto 3D